

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

B.E. in Robotics & Artificial Intelligence**Scheme of Teaching and Examinations 2022**

Outcome Based Education (OBE) and Choice Based Credit System (CBCS)

(Effective from the academic year 2023-24)

III SEMESTER

I SEMESTER													
Sl. No	Course	Course Code	Course Title	Teaching Department (TD) and Question and Answer Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing	SDA	Duration in hours	CIE Marks	SEE Marks	Total Marks	
					L	T	P	S					
1	PCC	BRI301	Fundamentals of Robotics & Applications	TD- RI/ME PSB -RI	3	0	0		03	50	50	100	3
2	IPCC	BRI302	Manufacturing Technology for Robots	TD: RI/ME PSB: RI	3	0	2		03	50	50	100	4
3	IPCC	BRI303	Analog and Digital Electronic Circuits	TD: ECE PSB: ECE	3	0	2		03	50	50	100	4
4	PCC	BRI304	Data Structures and Algorithms	TD: CSE PSB: CSE	3	0	0		03	50	50	100	3
5	PCCL	BMEL305	Introduction to Modeling and Design for Manufacturing	TD: ME/RI PSB: RI	0	0	2		03	50	50	100	1
6	ESC	BRI306X	ESC/ETC/PLC	TD: RI/ME/ECE/CSE PSB: Respective board	3	0	0		03	50	50	100	3
7	UHV	BSCK307	Social Connect and Responsibility	Any Department	0	0	2		01	100	---	100	1
8	AEC/ SEC	BRI358x	Ability Enhancement Course/Skill Enhancement Course – III	Any Department	If the course is a Theory				01	50	50	100	1
					1	0	0						
					If a course is a laboratory				03				
					0	0	2						
9	MC	BNSK359	National Service Scheme (NSS)	NSS coordinator	0	0	2			100	---	100	0
		BPEK359	Physical Education (PE) (Sports and Athletics)	Physical Education Director									
		BYOK359	Yoga	Yoga Teacher									
Total									550	350	900	20	

PCC: Professional Core Course, **PCCL:** Professional Core Course laboratory, **UHV:** Universal Human Value Course, **MC:** Mandatory Course (Non-credit), **AEC:** Ability Enhancement Course, **SEC:** Skill Enhancement Course, **L:** Lecture, **T:** Tutorial, **P:** Practical **S= SDA:** Skill Development Activity, **CIE:** Continuous Internal Evaluation, **SEE:** Semester End Evaluation. **K:** This letter in the course code indicates common to all the stream of engineering. **ESC:** Engineering Science Course, **ETC:** Emerging Technology Course, **PLC:** Programming Language Course

Engineering Science Course (ESC/ETC/PLC)(3:0:0:0) (Note- Student should opt for the course which should not be similar to the course opted in 1st Year)			
BRI306A	Material Science	BRI306C	Unmanned Aerial Vehicles (UAV)
BRI306B	Linear Integrated Circuits		
Ability Enhancement Course – III			
BRI358A	Data Structures and Algorithms Lab (0:0:2:0)	BRI358C	Fundamentals of Virtual Reality and App Development (1:0:0:0)
BRI358B	Applications of MAT LAB (0:0:2:0)	BRI358D	Introduction to C++ Programming (1:0:0:0)

Professional Core Course (IPCC): Refers to Professional Core Course Theory Integrated with practical of the same course. Credit for IPCC can be 04 and its Teaching– Learning hours (L: T: P) can be considered as (3: 0: 2) or (2: 2: 2). The theory part of the IPCC shall be evaluated both by CIE and SEE. The practical part shall be evaluated by only CIE (no SEE). However, questions from the practical part of IPCC shall be included in the SEE question paper. For more details, the regulation governing the Degree of Bachelor of Engineering /Technology (B.E./B.Tech.) 2022-23 may please be referred.

National Service Scheme /Physical Education/Yoga: All students have to register for any one of the courses namely National Service Scheme (NSS), Physical Education (PE)(Sports and Athletics), and Yoga(YOG) with the concerned coordinator of the course during the first week of III semesters. Activities shall be carried out between III semester to the VI semester (for 4 semesters). Successful completion of the registered course and requisite CIE score is mandatory for the award of the degree. The events shall be appropriately scheduled by the colleges and the same shall be reflected in the calendar prepared for the NSS, PE, and Yoga activities. These courses shall not be considered for vertical progression as well as for the calculation of SGPA and CGPA, but completion of the course is mandatory for the award of degree.

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IV SEMESTER													
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing	Self -Study	Duration in hours	CIE Marks	SEE Marks	Total Marks	
1	PCC	BRI401	Robot Kinematics, Dynamics and Control	TD: RI/ME PSB: RI	3	0	0		03	50	50	100	3
2	IPCC	BRI402	Mechanics and Measurement Systems for Robots	TD: RI/ME PSB:RI	3	0	2		03	50	50	100	4
3	IPCC	BRI403	Microcontroller	TD: ECE PSB: ECE	3	0	2		03	50	50	100	4
4	PCCL	BRIL404	Robot Programming & Simulation Lab	TD: RI/ME PSB: RI	0	0	2		03	50	50	100	1
5	ESC	BRI405x	ESC/ETC/PLC	TD: RI/ ME/ECE/CSE PSB: Respective board	3	0	0		03	50	50	100	3
6	AEC/ SEC	BRI456x	Ability Enhancement Course/Skill Enhancement Course- IV	Any Department	If the course is Theory				01	50	50	100	1
					1	0	0						
					If the course is a lab				03				
					0	0	2						
4	BSC	BBOK407	Biology For Engineers	TD / PSB: BT, CHE,	3	0	0		03	50	50	100	3
7	UHV	BUHK408	Universal human values course	Any Department	1	0	0		01	50	50	100	1
9	MC	BNSK459	National Service Scheme (NSS)	NSS coordinator	0	0	2			100	---	100	0
		BPEK459	Physical Education (PE) (Sports and Athletics)	Physical Education Director									
		BYOK459	Yoga	Yoga Teacher									
Total									500	400	900	20	

PCC: Professional Core Course, **PCCL:** Professional Core Course laboratory, **UHV:** Universal Human Value Course, **MC:** Mandatory Course (Non-credit), **AEC:** Ability Enhancement Course, **SEC:** Skill Enhancement Course, **L:** Lecture, **T:** Tutorial, **P:** Practical **S= SDA:** Skill Development Activity, **CIE:** Continuous Internal Evaluation, **SEE:** Semester End Evaluation. K: This letter in the course code indicates common to all the stream of engineering.

Ability Enhancement Course / Skill Enhancement Course - IV

BRI456A	Spread Sheets for Engineers (0:0:2:0)	BRI456C	Supervisory Control and Data Acquisition System (SCADA) (1:0:0:0)
BRI456B	Embedded C Basics (0:0:2:0)	BRI456D	Applications of Raspberry Pi Controllers (0:0:2:0)
Engineering Science Course (ESC/ETC/PLC) (3:0:0:0)(Note- Student should opt for the course which should not be similar to the course opted in 1st Year)			
BRI405A	Fuzzy Logic for Robotics	BRI405C	Robot Vision
BRI405B	Machine Learning Fundamentals		

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